



Beyond Prediction: Counterfactual Deliberation for L4 Autonomy

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DriveX Workshop @ ECCV | September 9th 2026

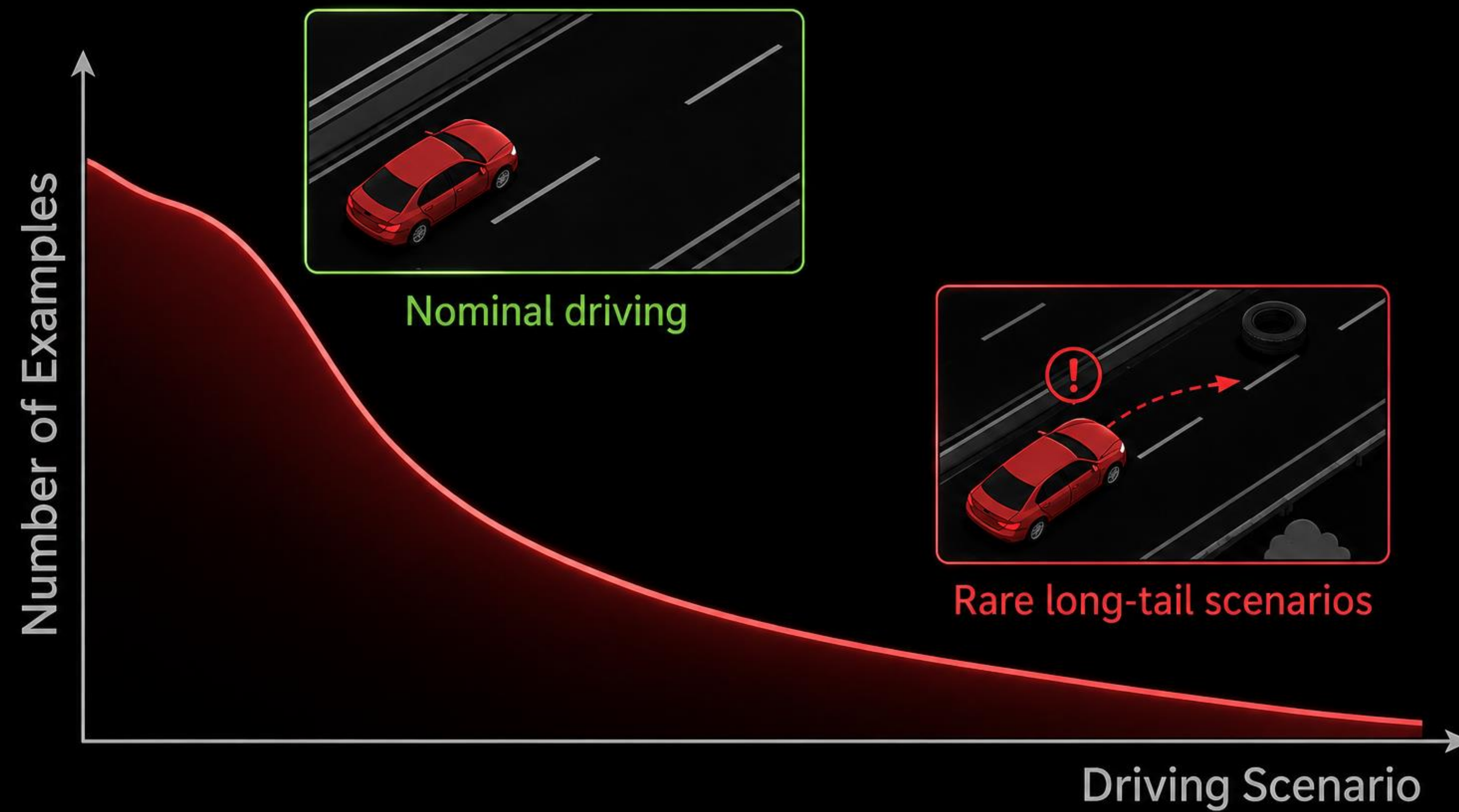


**“Training alone is
insufficient.”**

L4 autonomy = Training +
Real-time counterfactual
decision-making under
uncertainty.”

A World of Possibilities

L4 autonomy must handle the long-tail: safely, everywhere, every day



Real-world data scarcity

World Model as Data Engines

Better Environments for Training Autonomous Vehicles

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World Model as Data Engines

Better Environments for Training Autonomous Vehicles

NVIDIA's World Foundation model to generate physics-accurate training data



Better Data

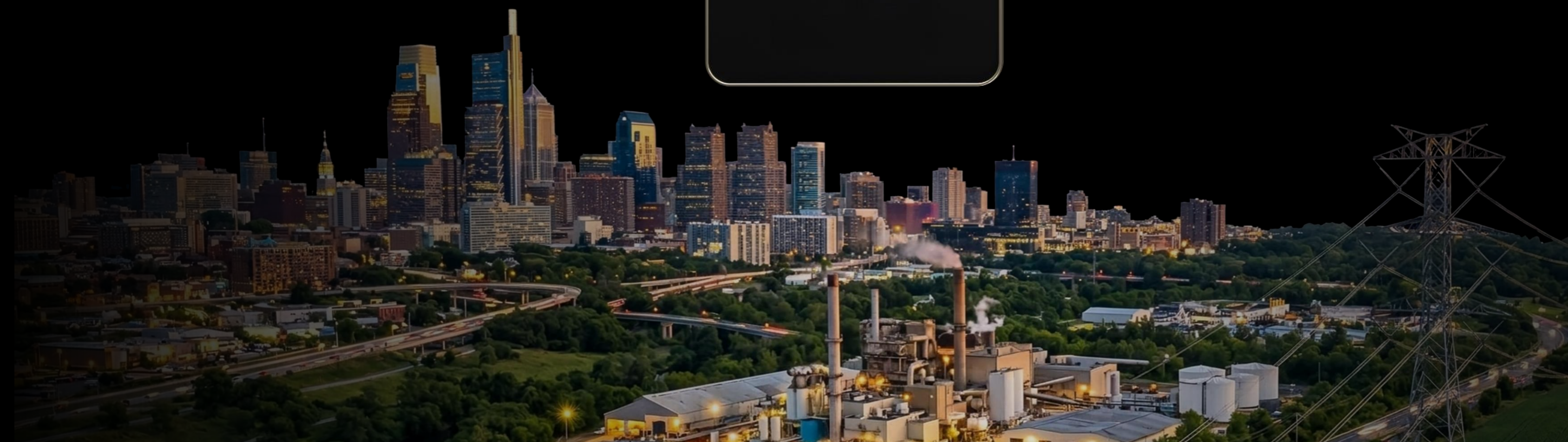
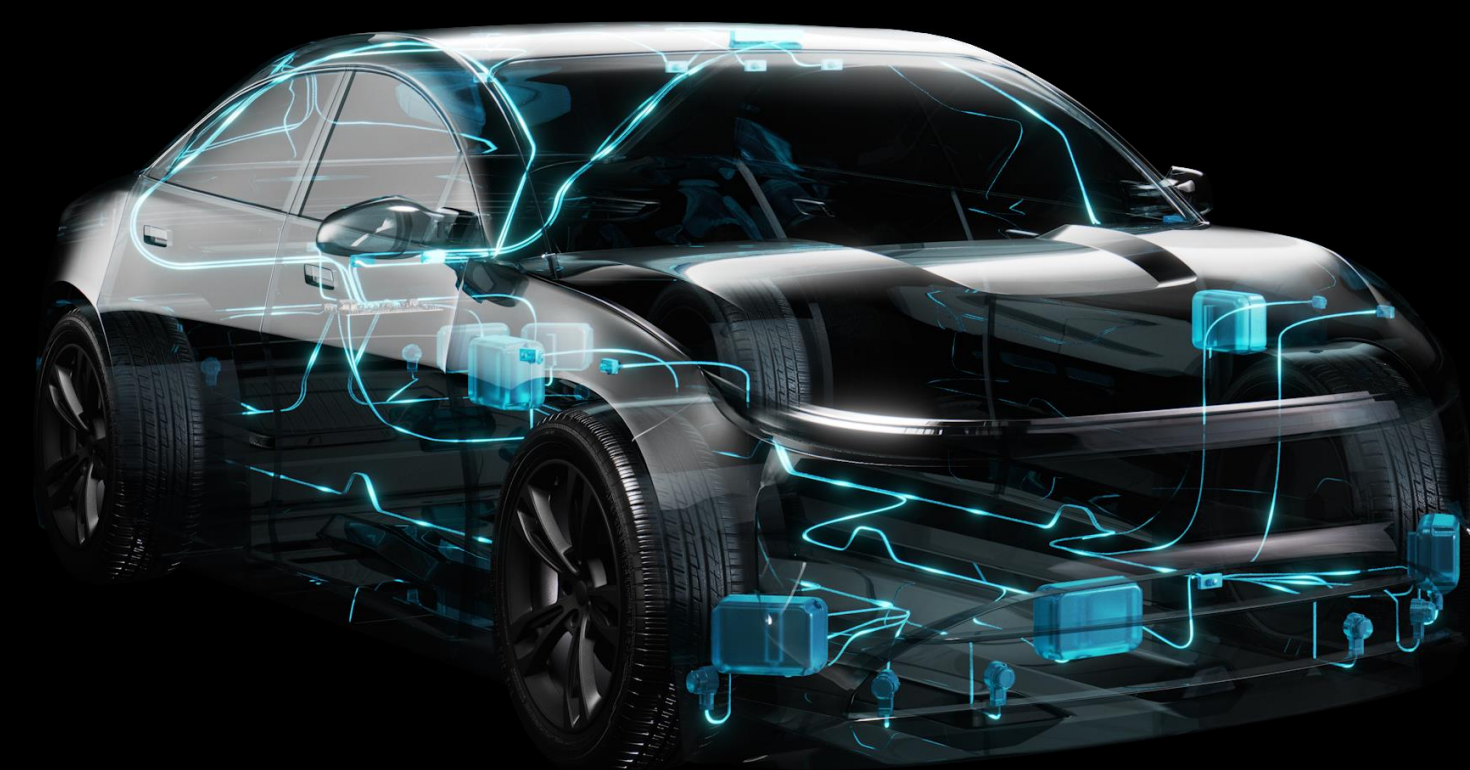


Better Environment



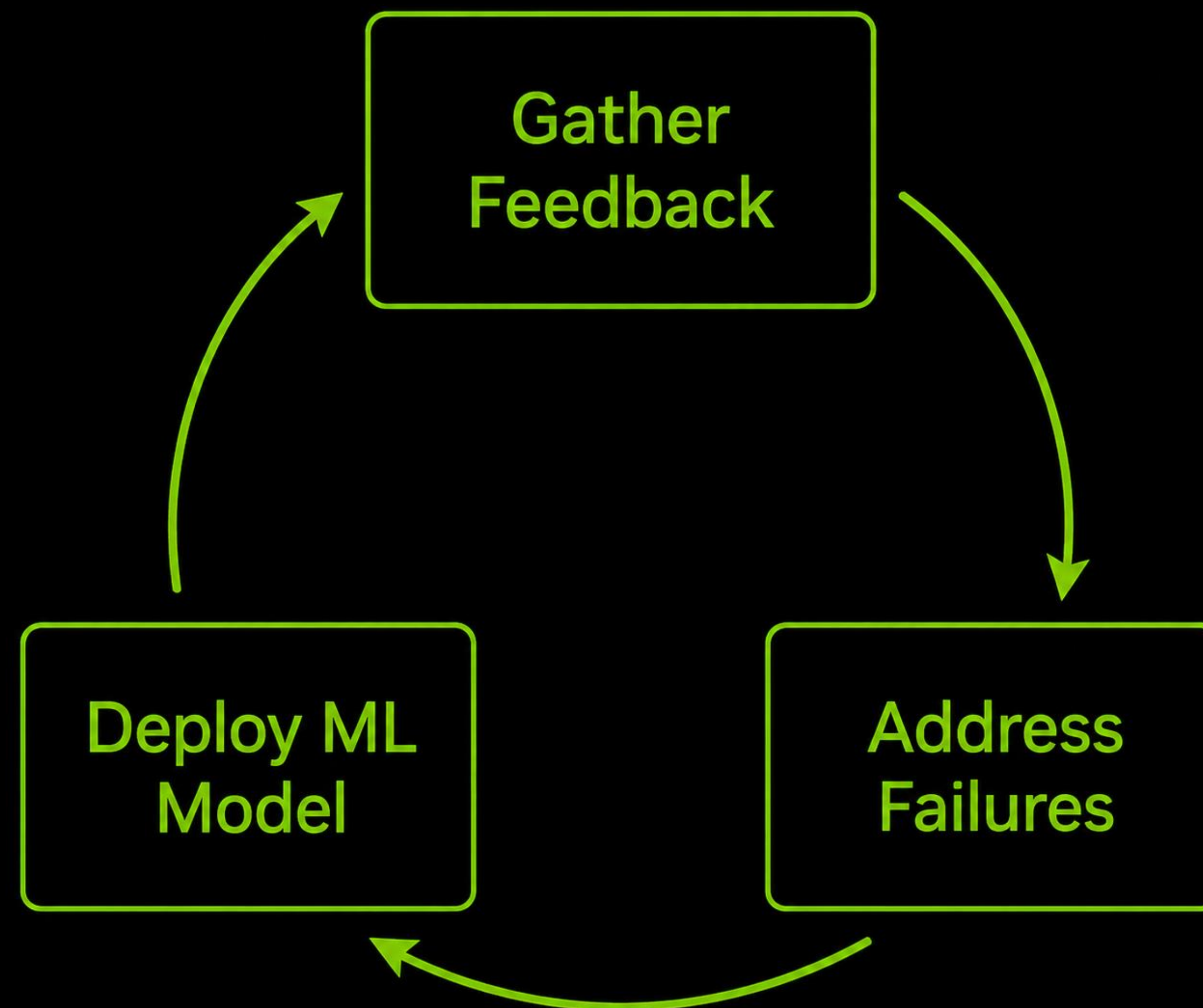
Better Starting Point

Cosmos



Addressing long-tail scenarios via the reactive AI flywheel

Wait for failures and fix them as they come through the flywheel



- Observe a failure
- Find similar cases
- Patch the incident
- Repeat when a new variant appears

Reactive flywheels do not scale to open-ended autonomy

From Reactive to Proactive Flywheels

A new paradigm for building generalizable AI systems

- Map feedback to task conditions
- Identify weakness patterns
- Route to data or model solutions
- Prevent related future failures



Don't just react to the past. Reason about the space of failures

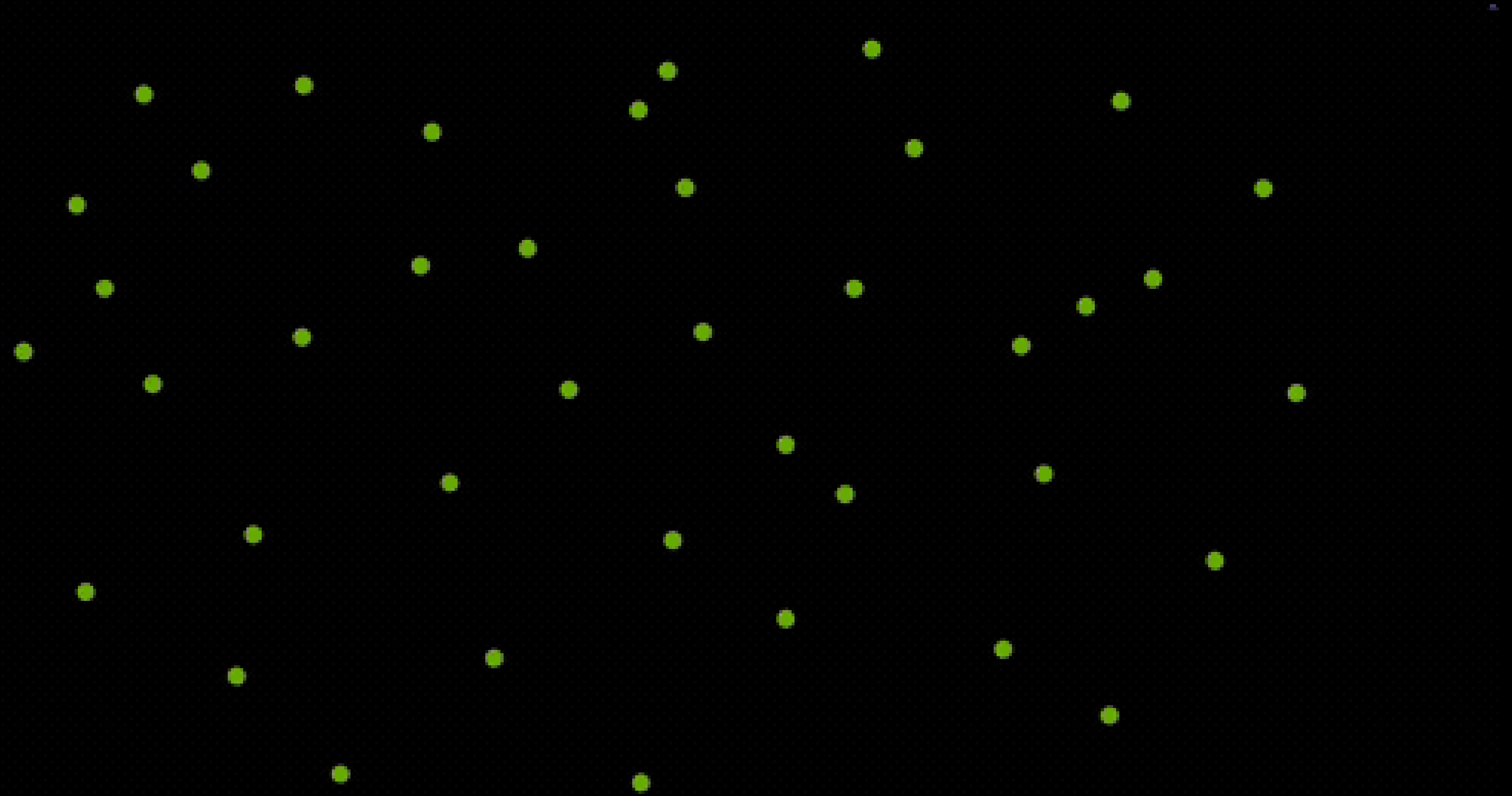
A Test-Space for Proactive Development

Map feedback to task conditions

The feature space that contains all situations, conditions, and facts the Physical AI model must know about to be deployed reliable in the real world

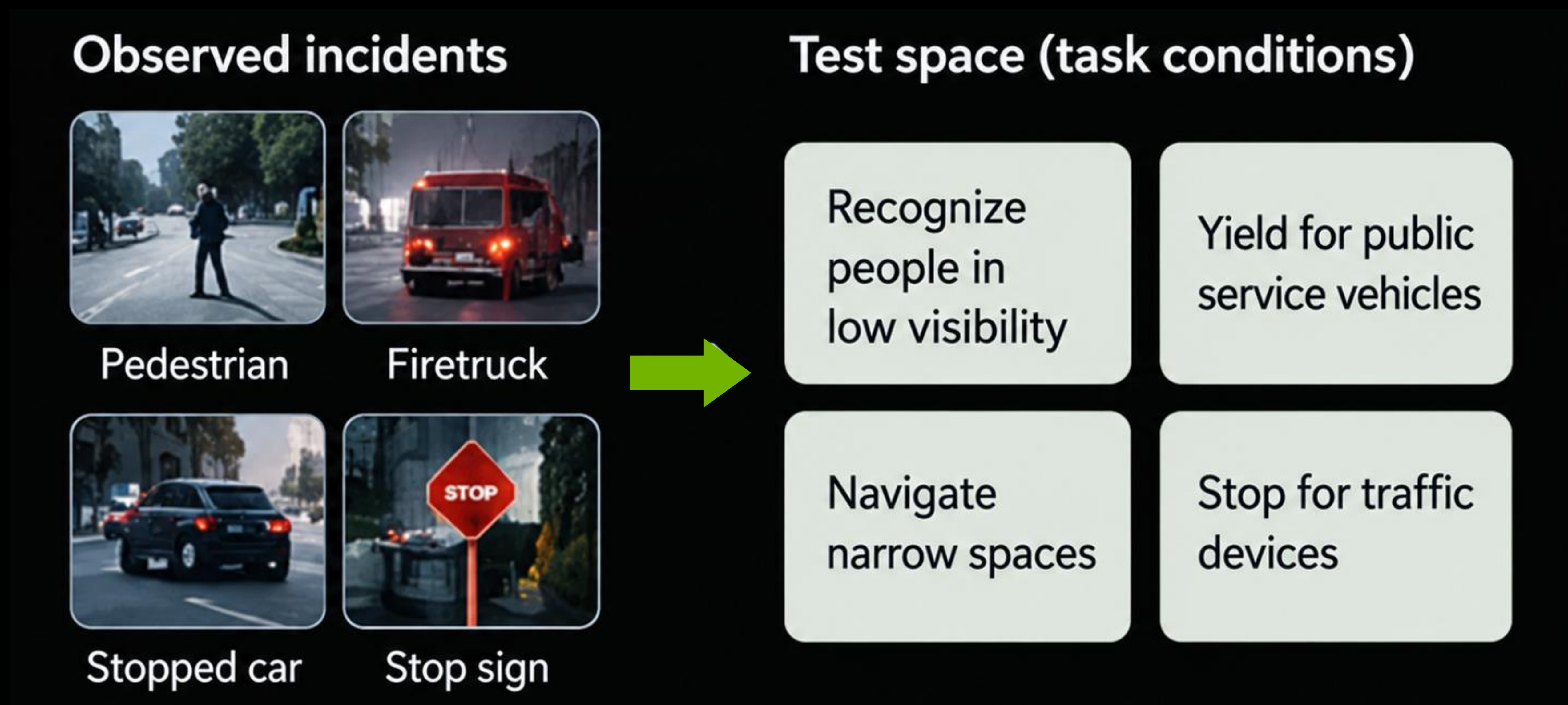
A Test-Space for Proactive Development

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A Test-Space for Proactive Development

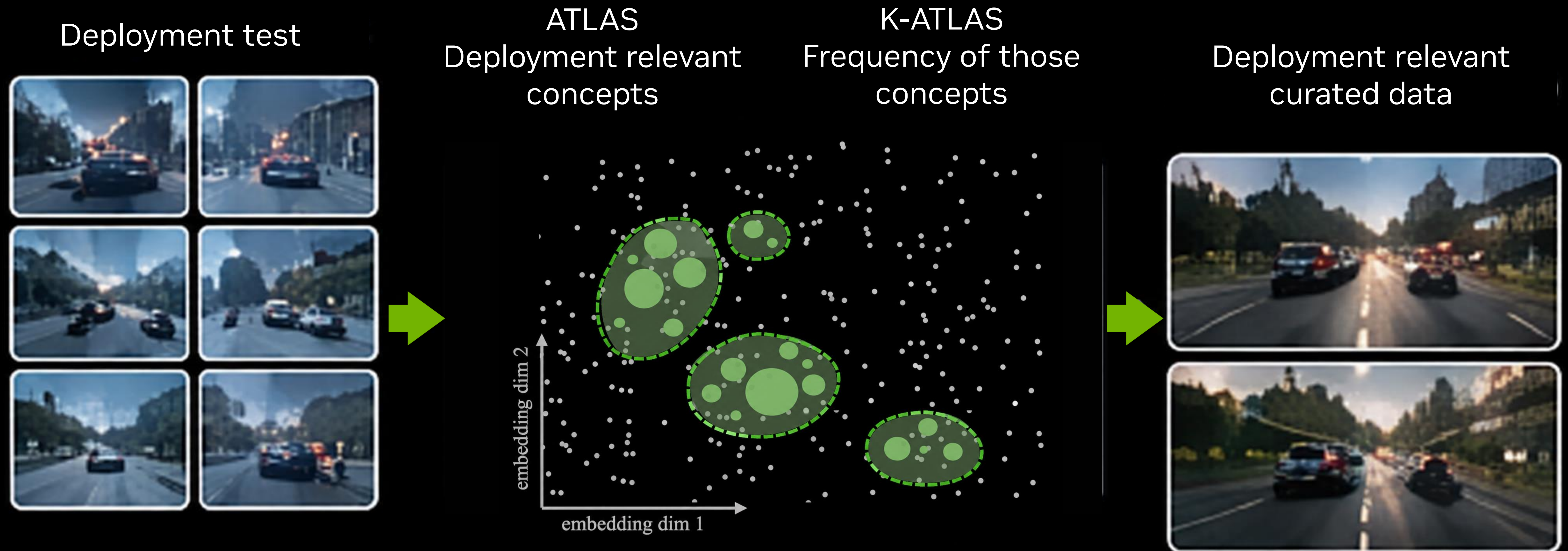
The feature space that contains all situations, conditions, and facts the Physical AI model must know about to be deployed reliable in the real world



From isolated fixes to systematic capability improvement.

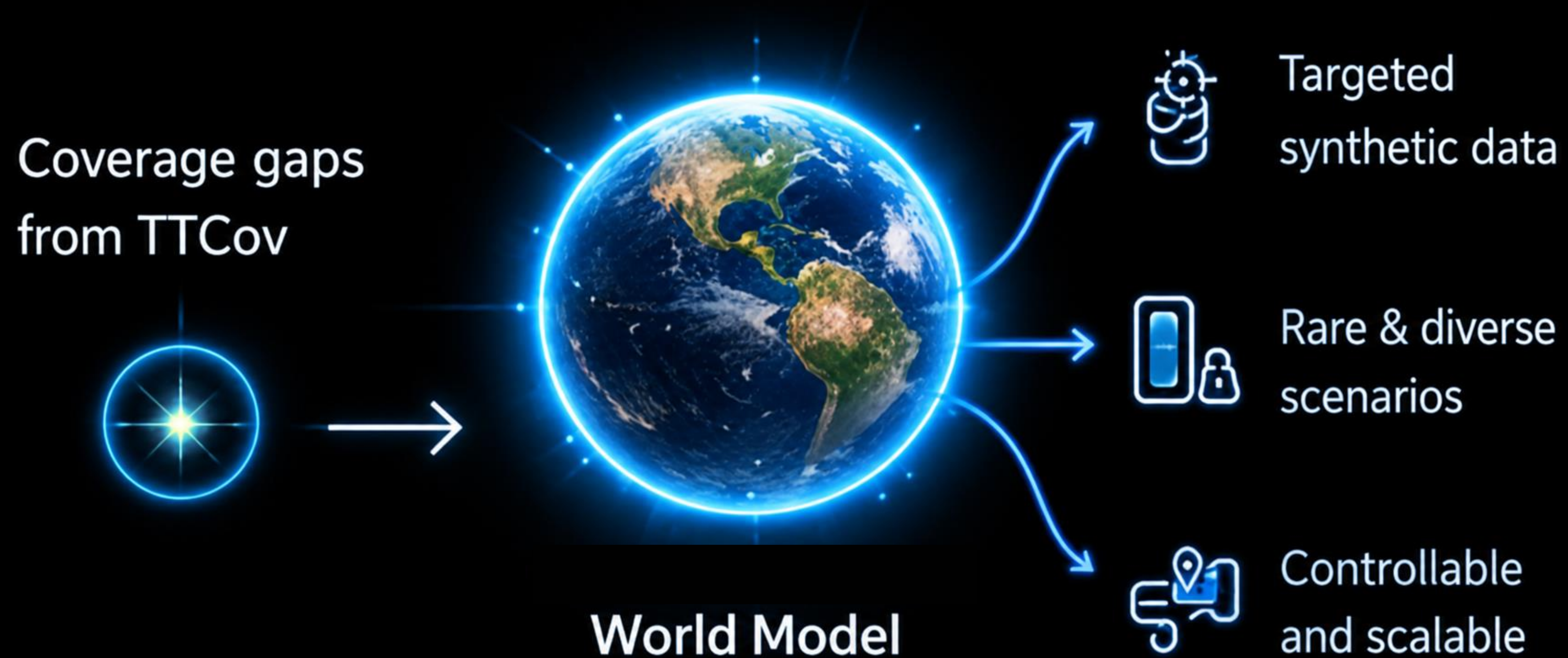
TTCOV: Test-Driven Data Curation

Use deployment conditions to determine what training data do we need



TTCOV + World Model as Data Engines

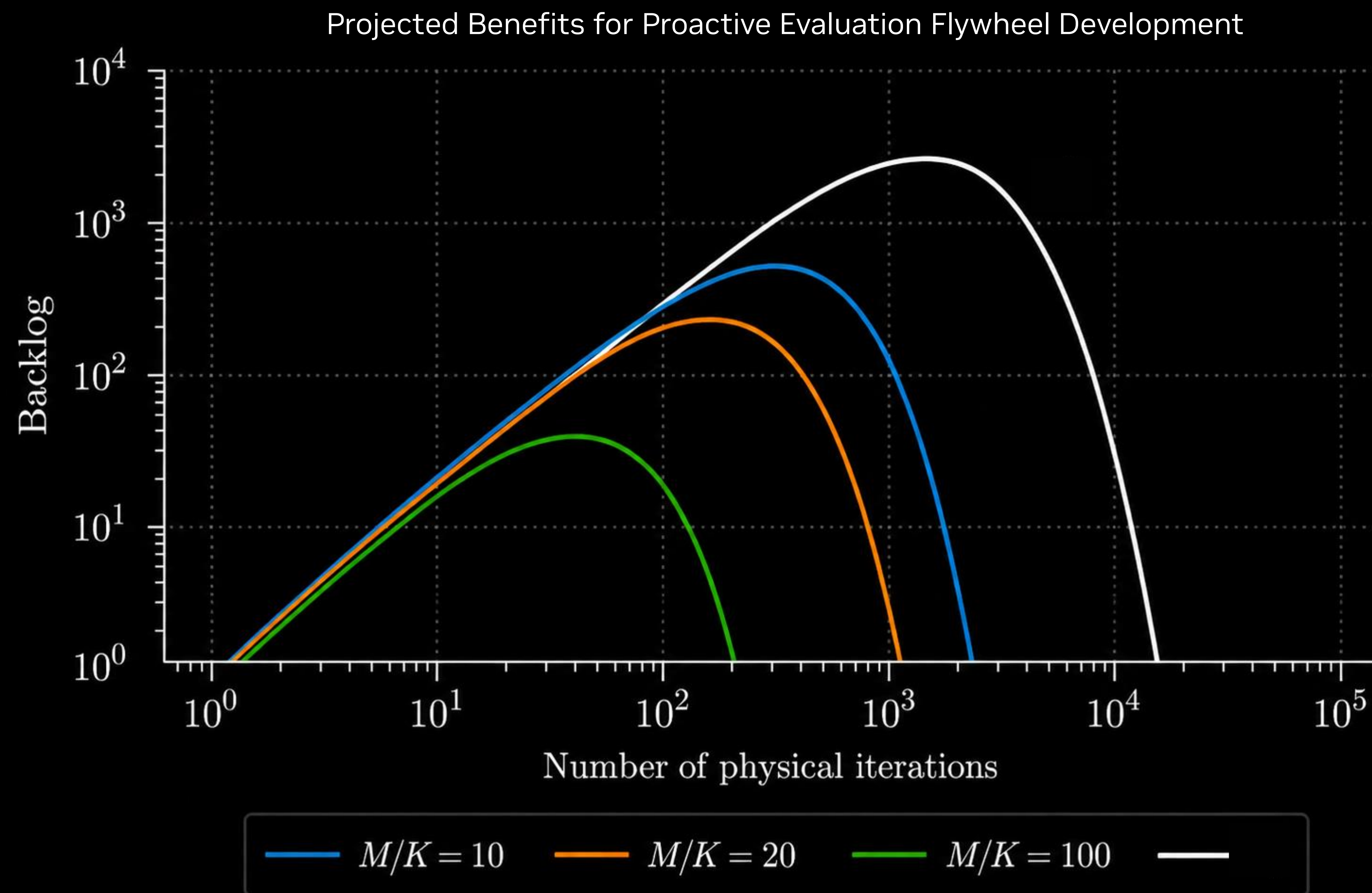
Generate the data the system needs, not just more data



From data scarcity to unlimited, purposeful data generation

From Reactive to Proactive Flywheels

Fewer iterations. Broader generalization



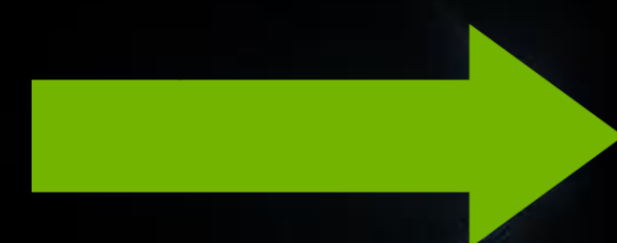
The backlog from reactive correction is an order of magnitude larger than from proactive correction

Real-World is Still Unpredictable

No amount of offline data removes the need to make the right decision now



Massive real + synthetic data
(ever-growing)



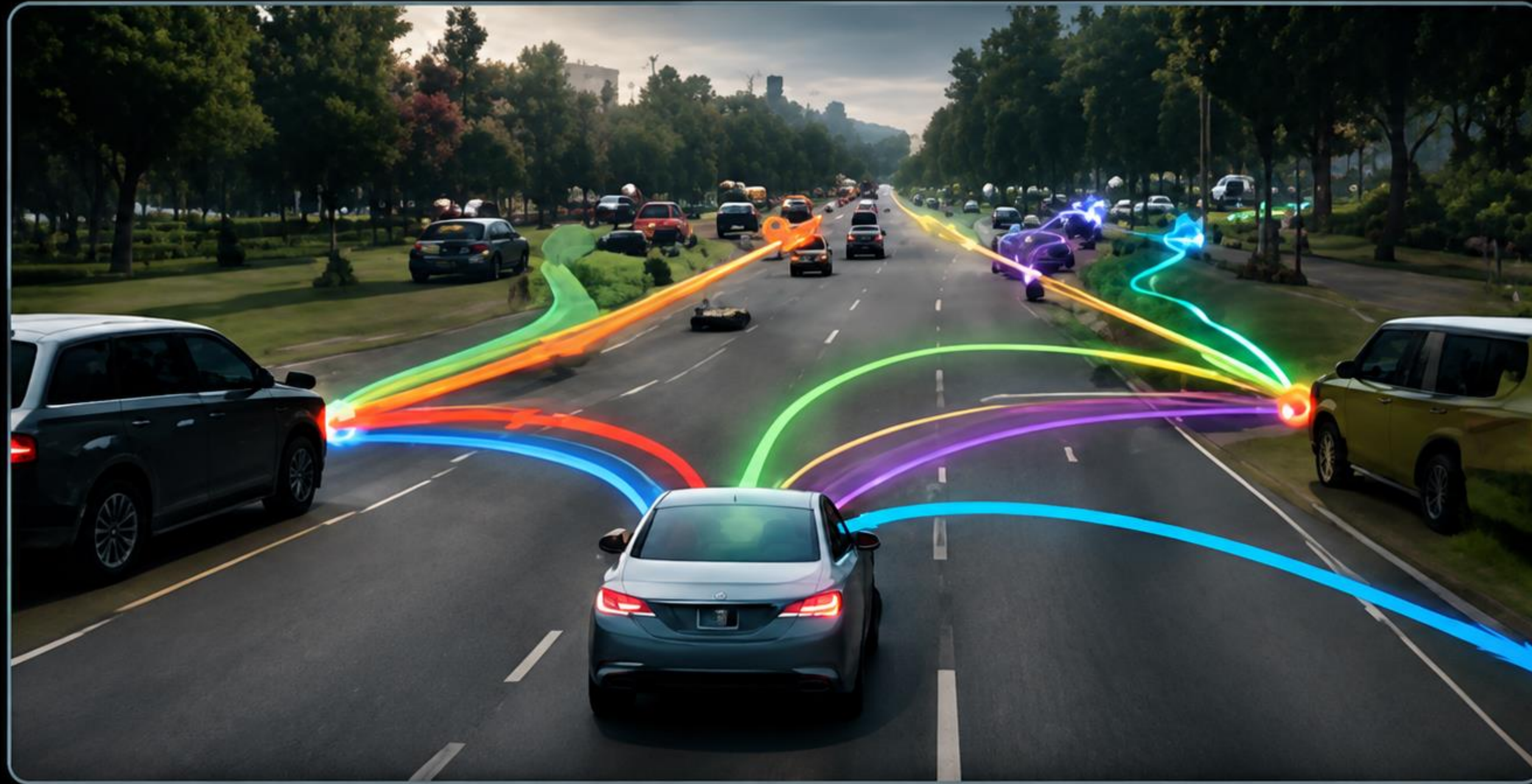
One specific, ambiguous situation.
What do we do now?

Training gets us far.

Training along is
insufficient.

At decision time: Counterfactual Deliberation

Given several feasible possibilities, imagine their consequences, evaluate and act safely



Action A
Proceed



Action B
Yield



Action C
Change lane

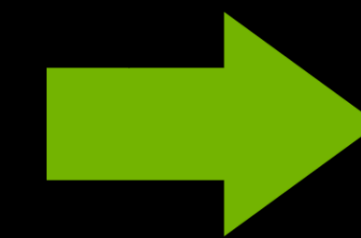
Think ahead. Act with greater confidence.

From Proactive Learning to Proactive Decision-Making

Test-time compute

Don't wait for the
failure.

Proactive development.



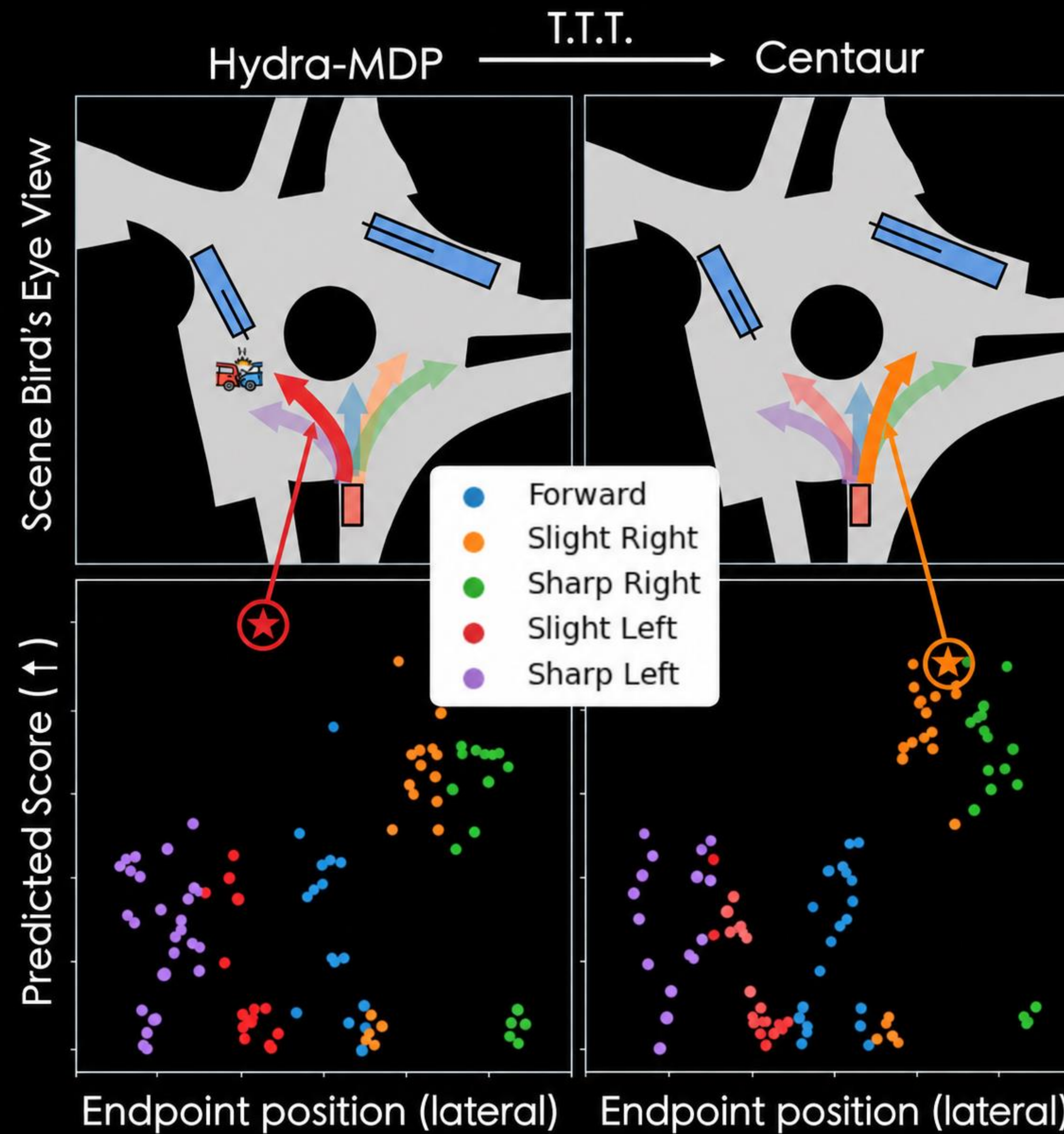
Don't wait for the
consequence.

Counterfactual deliberation

Same philosophy. Different timescales.

Test-Time Training: Think again when it matters

Test-time adaptation for out-of-distribution situations.



Detect uncertainty



Adapt at test-time



Improved robustness

Counterfactual Deliberation under Uncertainty

Think ahead before we act

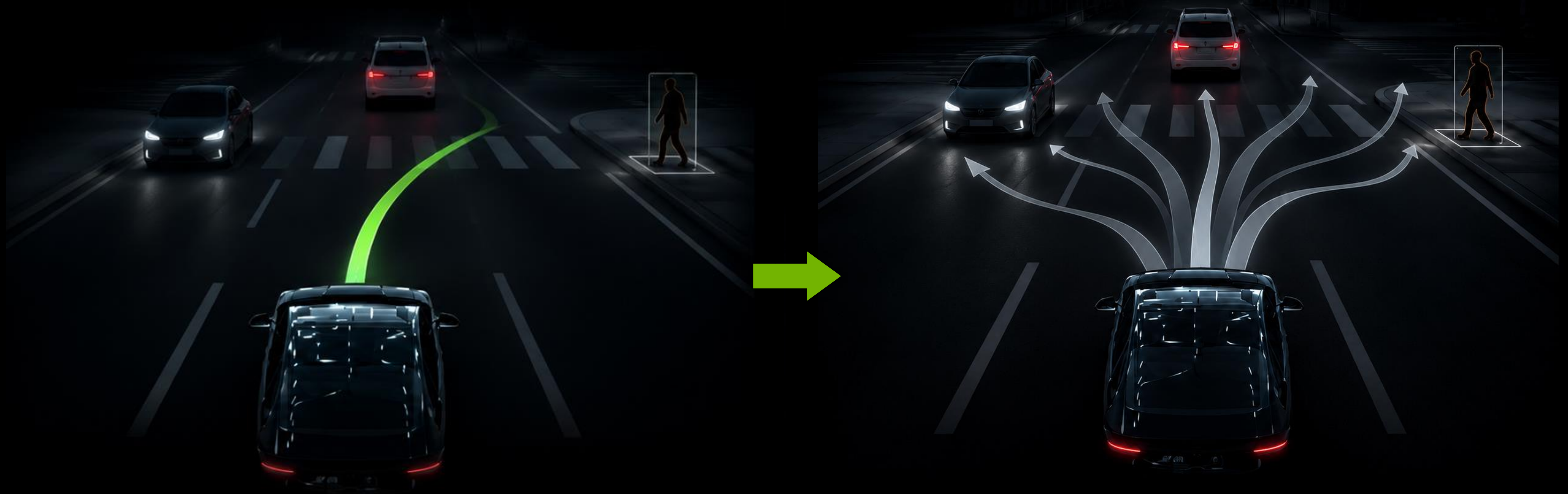


Reason over multiple plausible futures, for all agents.

Hydra-Series

Hydra Series: Multimodal Planners

Hydra models produce multiple feasible trajectories



Single Prediction
One trajectory, one future.

Multimodal Prediction
Multiple plausible futures.

Hydra-like planners move safety decisions toward inference

From structured trajectory scoring to deliberation over feasible futures

1 STRUCTURE THE ACTION SPACE

Hydra-MDP++

Distills rule-planner expertise into separate scores for safety, compliance, progress and comfort.

Hydra-NeXt

Adds short-horizon control prediction and kinematic refinement to **improve closed-loop execution**.

DriveSuprim

Sharpens ranking with **coarse-to-fine filtering**, hard negatives, targeted augmentation and self-distillation.

2 LEARN FROM DRIVING OBJECTIVES

ZTRS

Uses **exhaustive reward supervision** over a discrete trajectory vocabulary, removing dependence on human demonstrations.

HAD

Scales reward-driven planning through physically structured diffusion, metric-decoupled RL and reward caching.

At ECCV 2026 – Poster session Saturday Afternoon

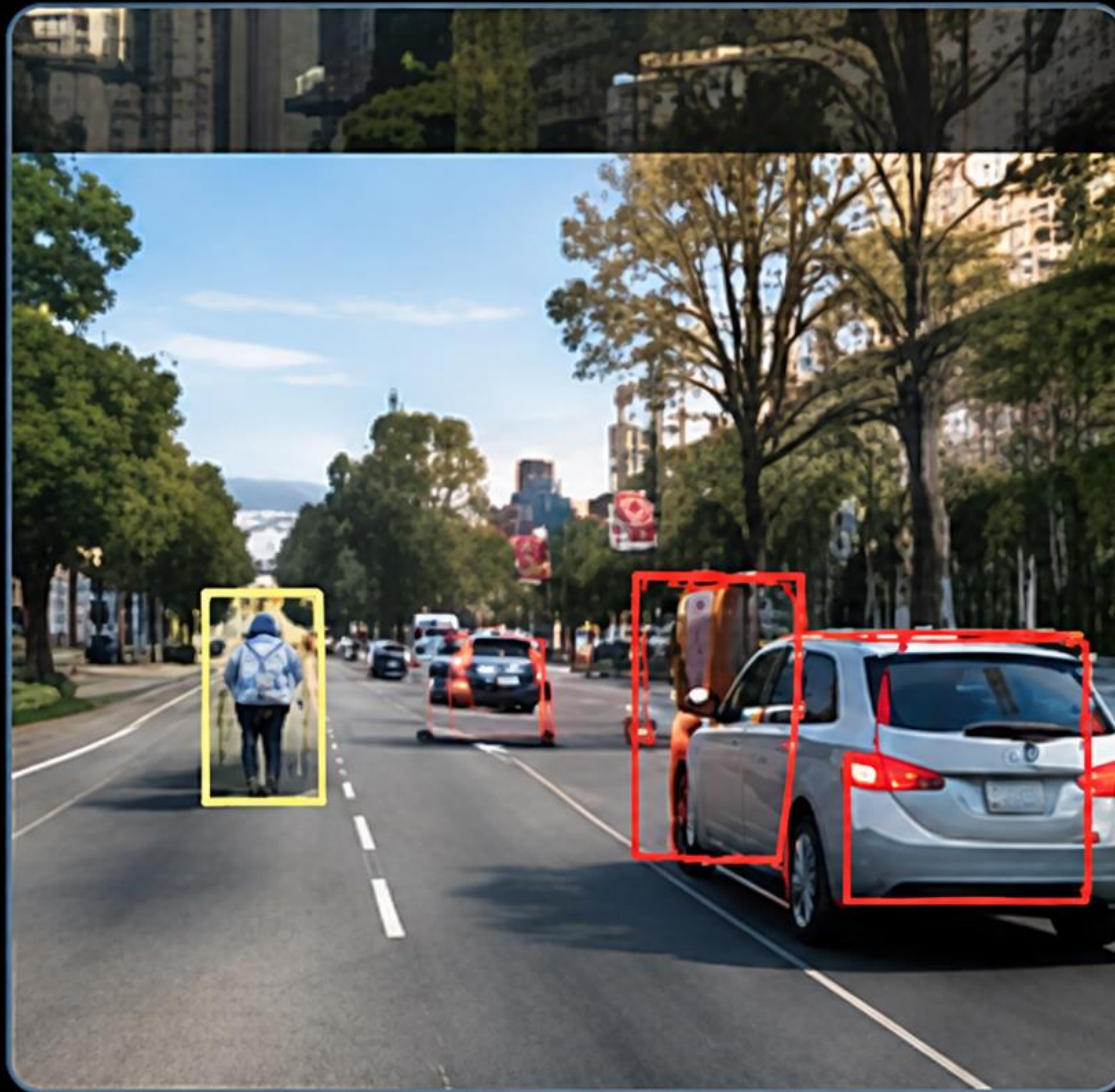
3 DELIBERATE IN THE OBSERVED SCENE

LDiP

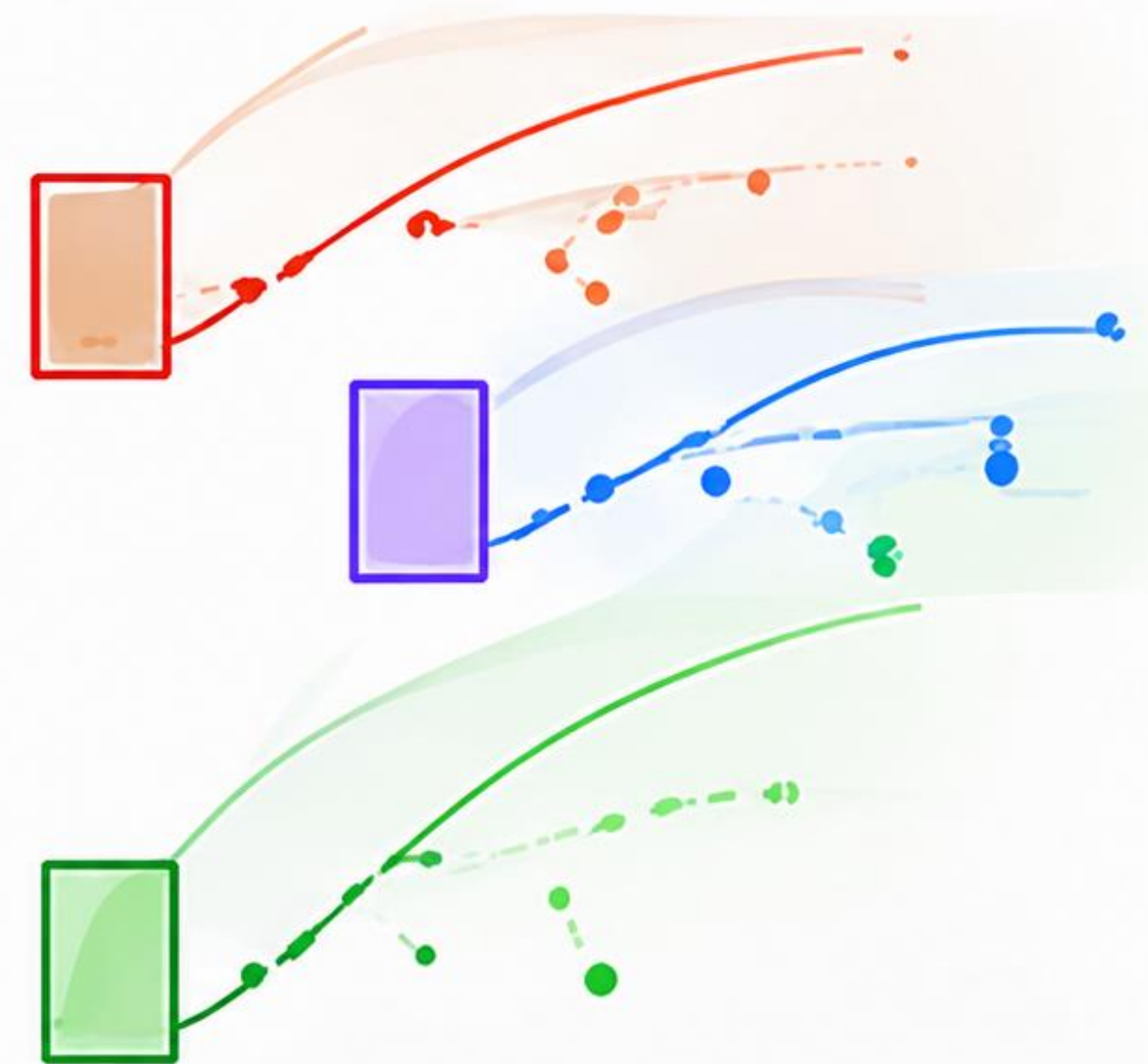
Iteratively scores, prunes and rescores a large set of physically feasible **trajectories at inference time**.

Hydra Series: Deliberation with Predicted Futures

Hydra uses a simple model to predict where the other objects will be at inference time



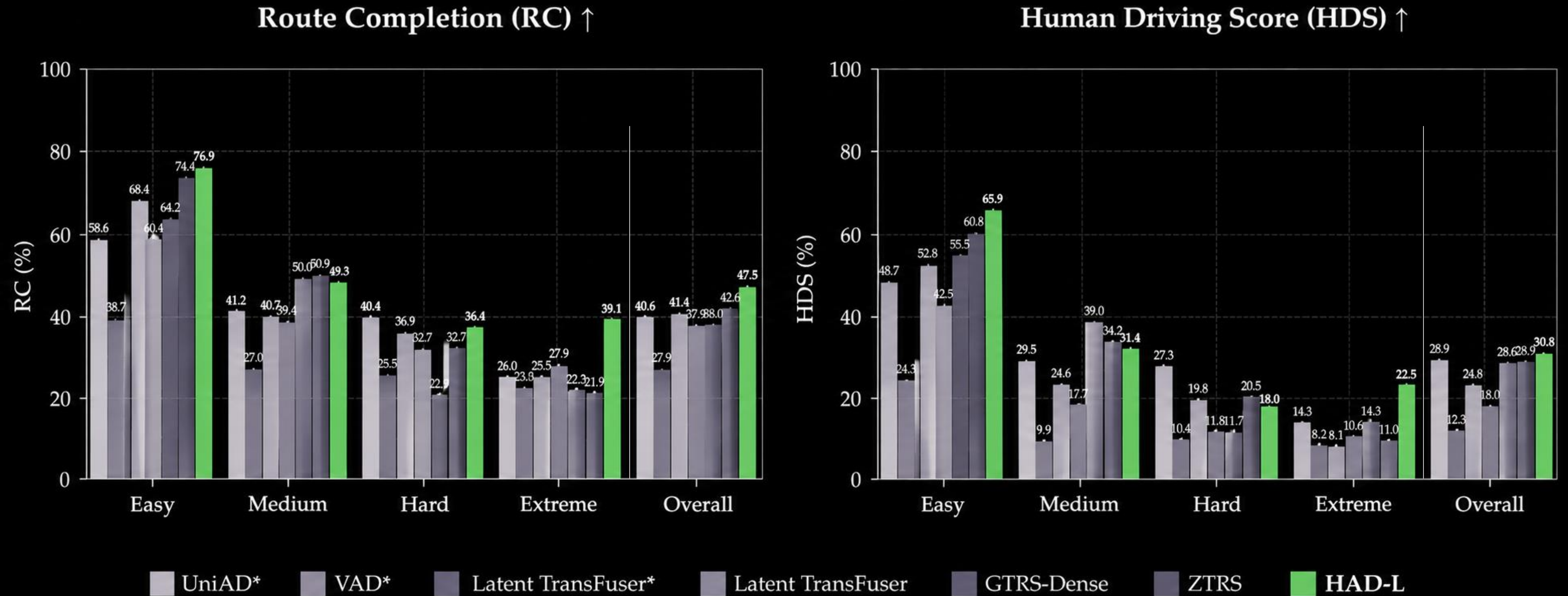
Simple future prediction model
(at inference time)



A simple model to dream about the future

Deliberation with Predicted Futures

HAD: Combining Hierarchical Diffusion with Metric-Decoupled RL for End-to-End Driving



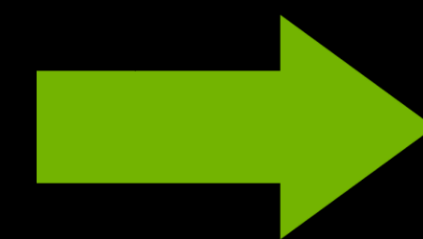
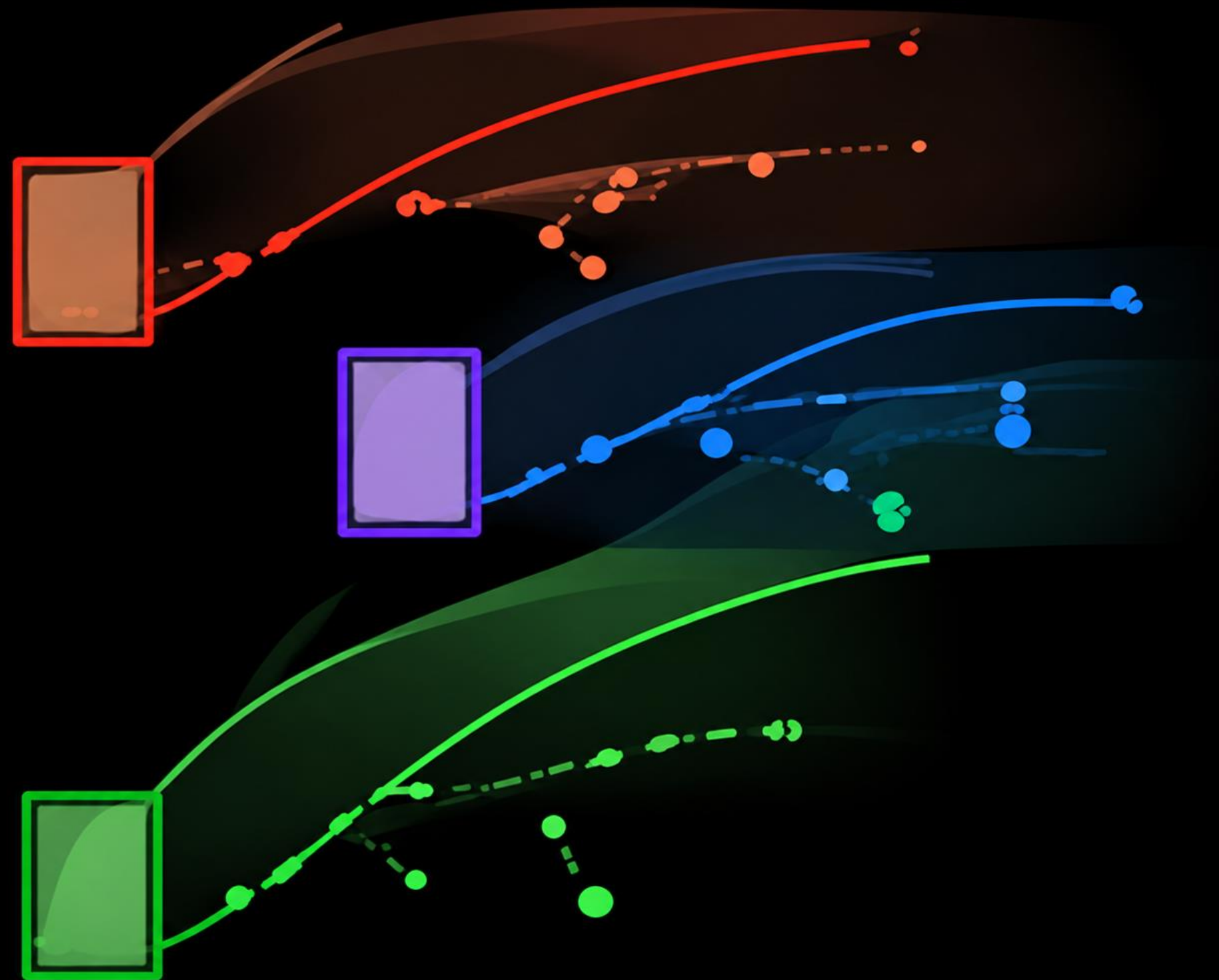
High-fidelity closed-loop benchmark HUGSIM

What is Next?

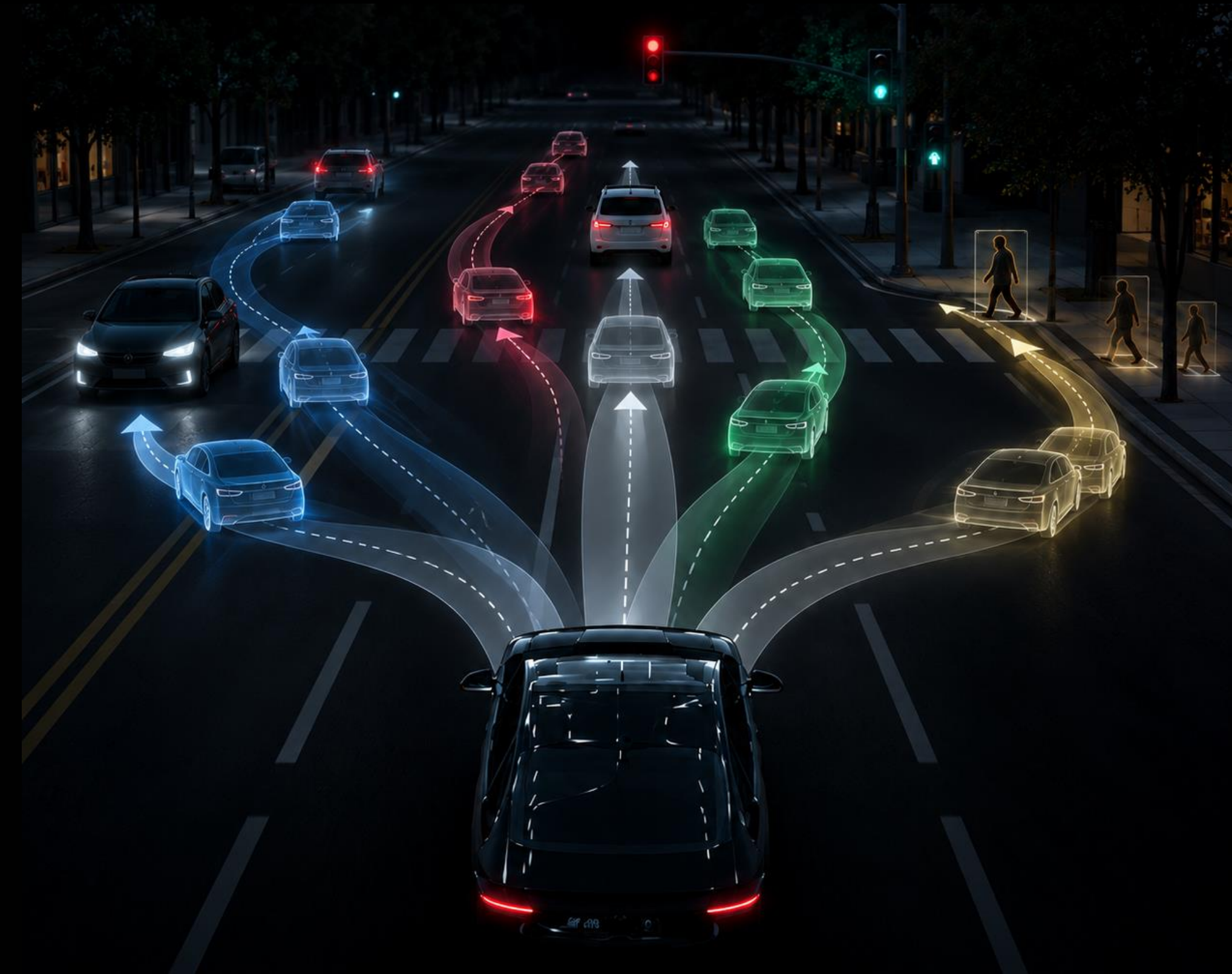
What is next? World-model-based deliberation

Richer, action-conditioned world models for interactive, multi-agent futures.

Simpler
Predict object motion



World-Model-Based
Simulate rich interactive worlds

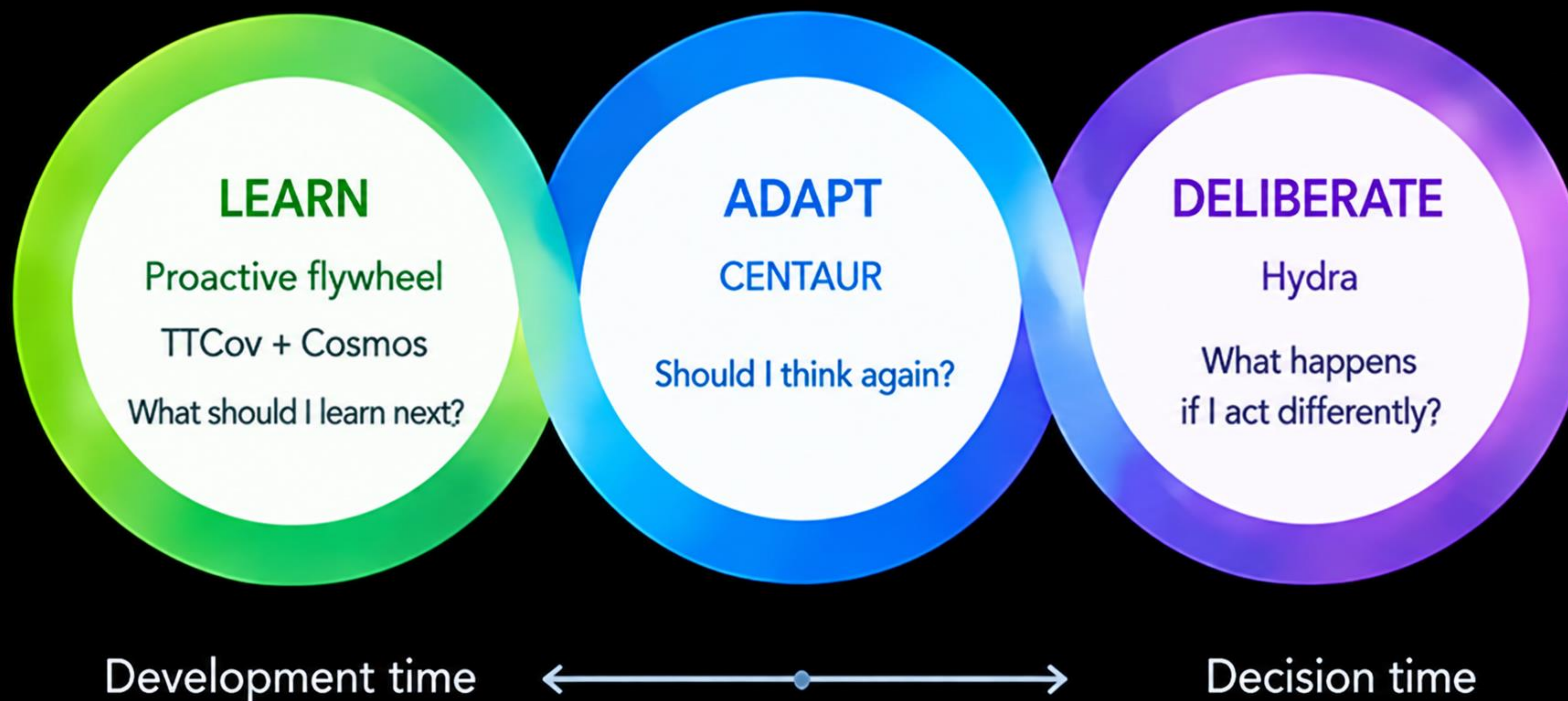


From simple prediction to rich, counterfactual imagination.

Take home message

Learn, Adapt, Deliberate

A complete stack for safe L4 autonomy



Safety through anticipation, across the entire lifecycle.

*Training builds experience.
Test-time deliberation turns
experience into judgment.*

Thank you!



